- why you follow the same patterns at work each day? Routine ??
- Learning is a relatively lasting change in behavior resulting from different techniques
- Learning refers to the process of acquiring new information, knowledge, behaviors, skills, values, or preferences.
- It is a complex and dynamic activity involving multiple cognitive, emotional, and contextual factors
- a theoretical orientation that emphasizes the study of observable behaviors and the role of environmental stimuli in shaping them. Pioneered by figures such as John B. Watson and B.F. Skinner



Major contribution in behavioral model

Ivan Pavlov

- Classical Conditioning

B.F. Skinner

- Operant Conditioning

Albert Bandura

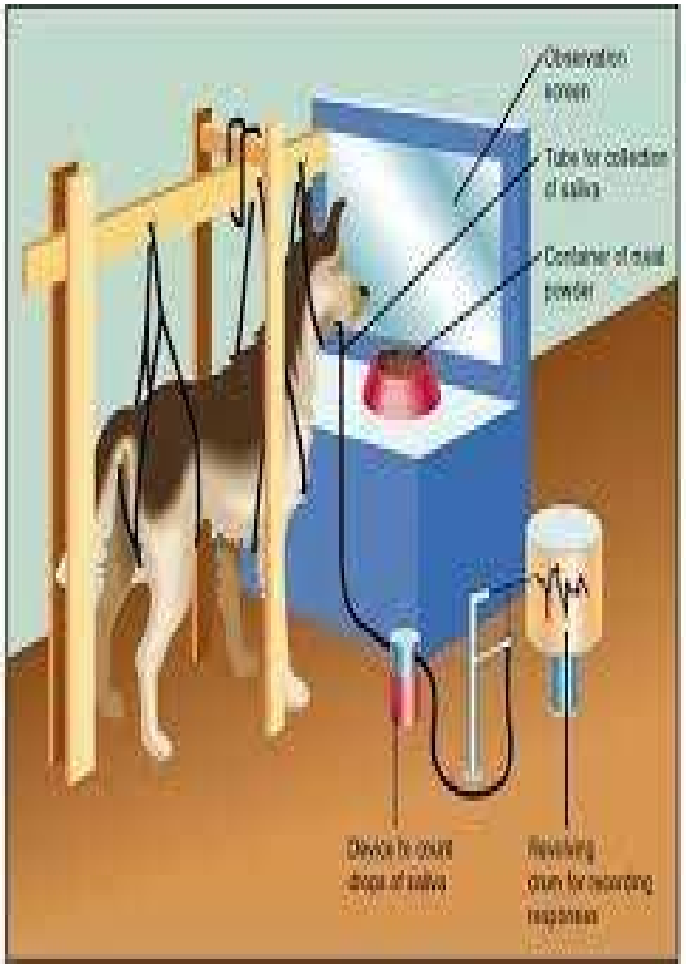
- Social Learning theory

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CLASSICAL CONDITIONING

- During the 1890s, Russian physiologist, Ivan Pavlov was researching salivation in dogs in response to being fed. He inserted a small test tube into the cheek of each dog to measure saliva when the dogs were fed (with a powder made from meat).
- When two events repeatedly occur close together in time, they become fused in a person's mind, and before long the person responds in the same way to both events.

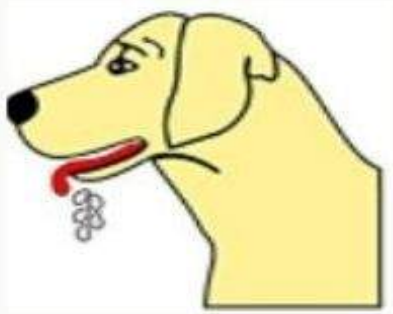
TEMPORAL ASSOCIATION



Before Conditioning



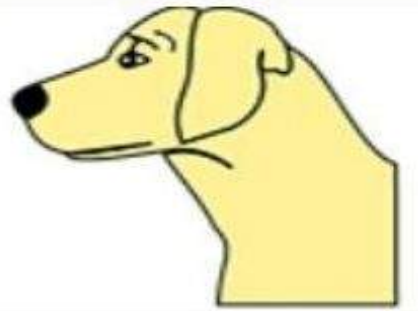
Unconditioned Stimulus



Unconditioned Response (Salivation)



Neutral Stimulus

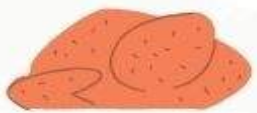


No Response

During Conditioning



+



Conditioned Stimulus



Unconditioned Response

After Conditioning

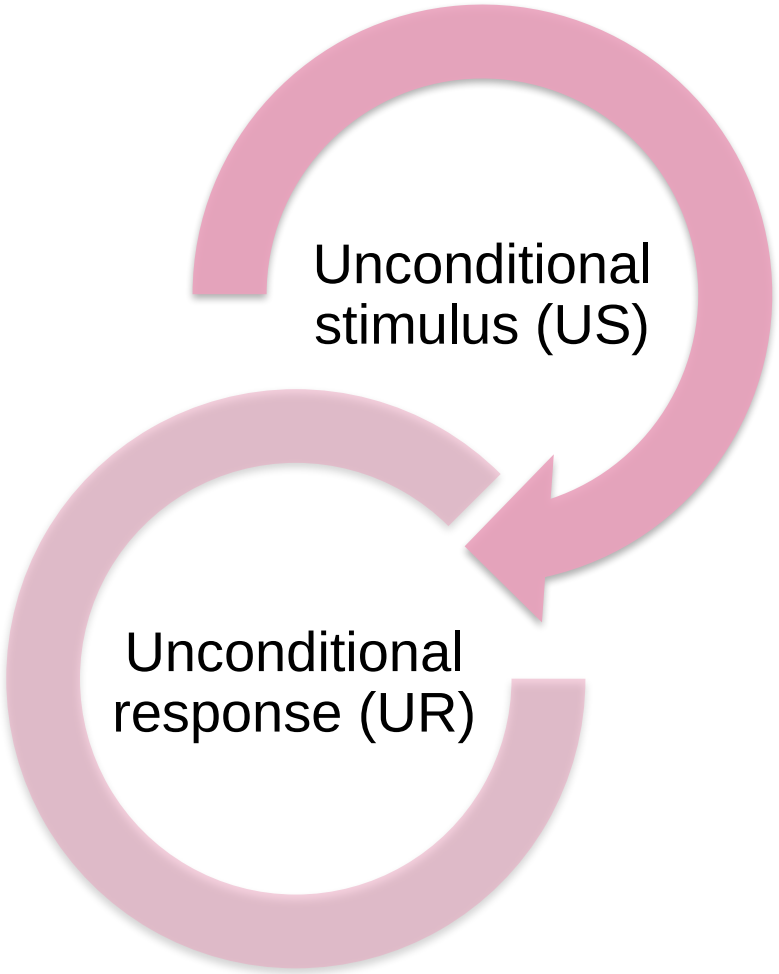


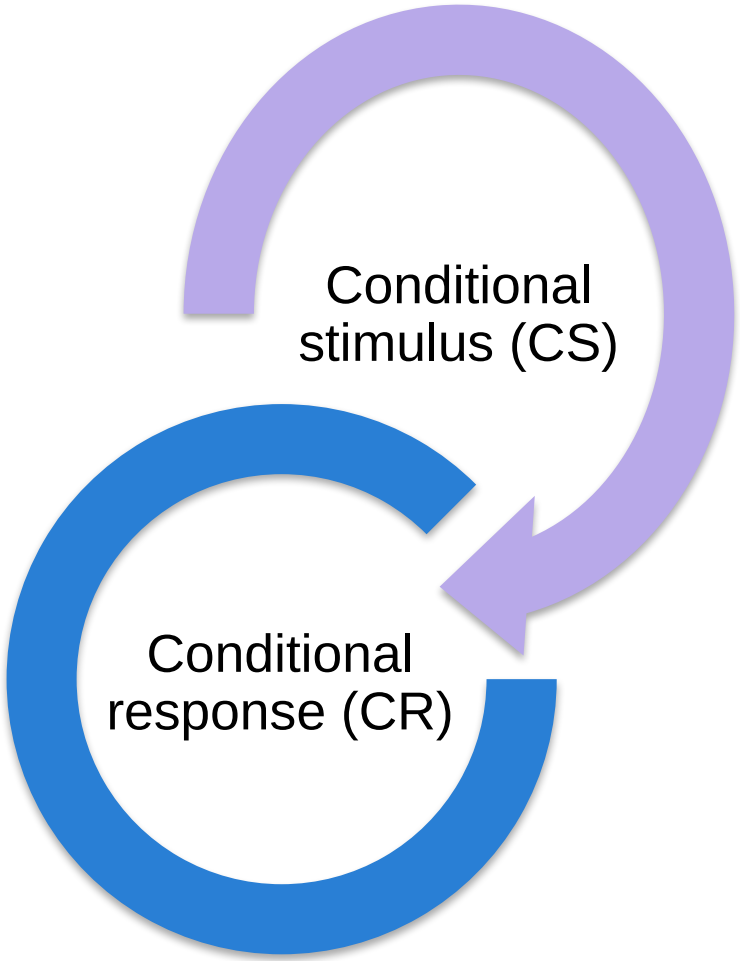
Conditioned Stimulus



Conditioned Response

tyonote


Unconditional stimulus (US)
Unconditional response (UR)


Conditional stimulus (CS)
Conditional response (CR)


EXAMPLE

OPERANT CONDITIONING

Operant conditioning, sometimes referred to as instrumental conditioning, is a method of learning that employs rewards and punishments for behavior. Through operant conditioning, an association is made between a behavior and a consequence (whether negative or positive) for that behavior.

- SKINNER INTRODUCED TWO TYPES OF BEHAVIORS
- Respondent (automatic/ reflexes)
- Operant (learned behaviors)



Components of classical conditioning

Reinforcement (INCREASE)

- Positive
- Negative

Punishment (DECREASE)

- Positive
- Negative

Reinforcement

- Any stimulus that ***increases the likelihood of the behavior*** is known as reinforcement

Positive Reinforcement

- When any reward or stimulus is ***added*** to the environment



Negative Reinforcement

- When any reward or stimulus is ***removed*** from th



Punishment

- Punishment is the presentation of an *adverse event* or outcome that causes a decrease in the behavior

Positive Punishment

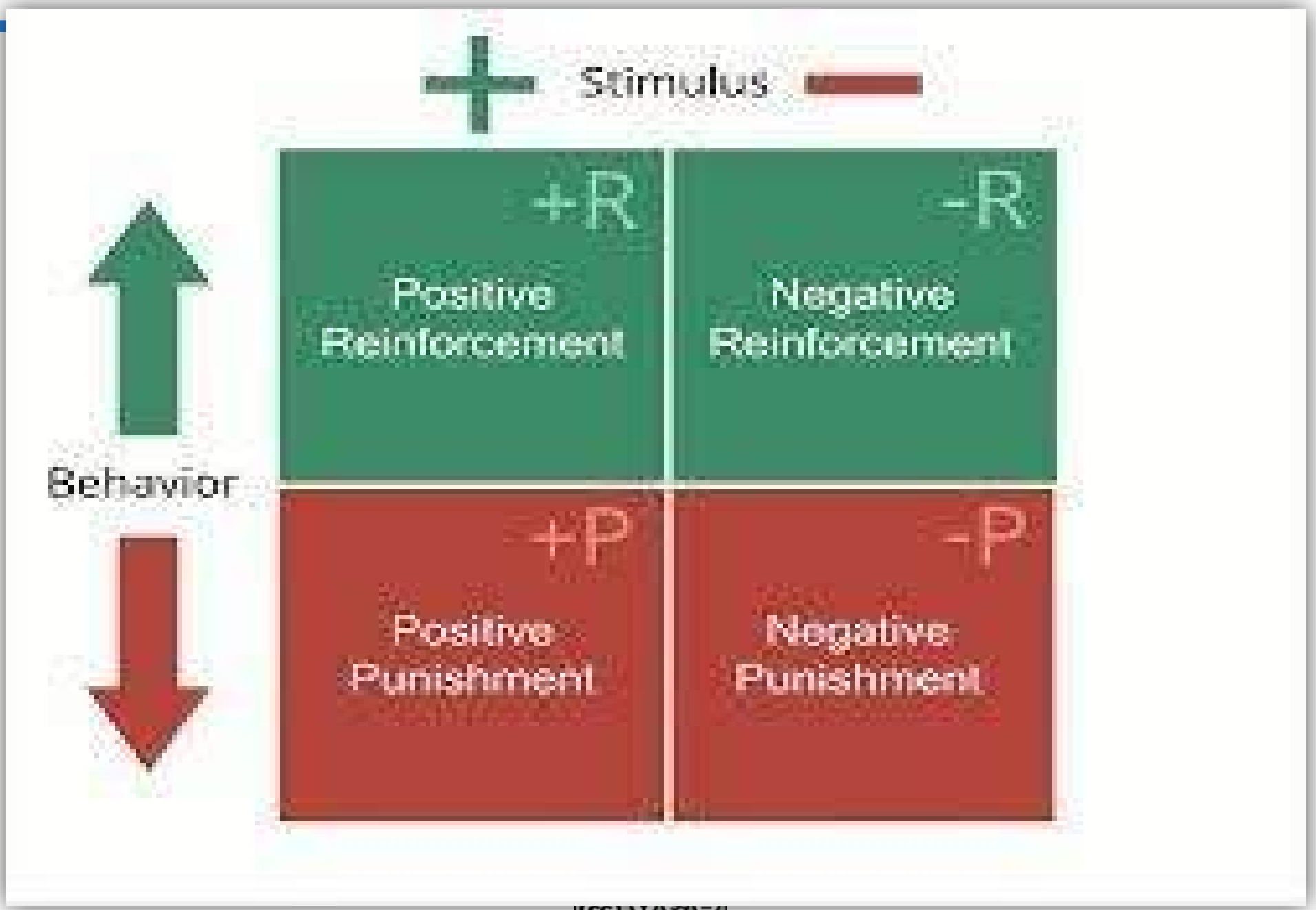
- Positive Punishment is when an unfavorable stimulus/consequence is ***added*** to the environment

Negative Punishment

- Negative Punishment is when the favorable event or outcome is ***removed*** from the environment







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Social Learning Theory

- In social learning theory, ***Albert Bandura (1977***) agrees with the behaviorist learning theories of ***classical conditioning*** and ***operant conditioning***.
- However, he adds two important ideas
  - Mediating processes occur between stimuli & responses.
  - Behavior is learned from the environment through the process of observational learning.

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- Children pay attention to some of these ***people (models)*** and ***encode*** their behavior.
- At a later time they may ***imitate (i.e., copy)*** the behavior they have observed.
- They may do this regardless of whether the behavior is ‘gender appropriate or not, but there are a number of processes that make it more likely that a child will reproduce the behavior that its society deems appropriate for its gender.



Observe

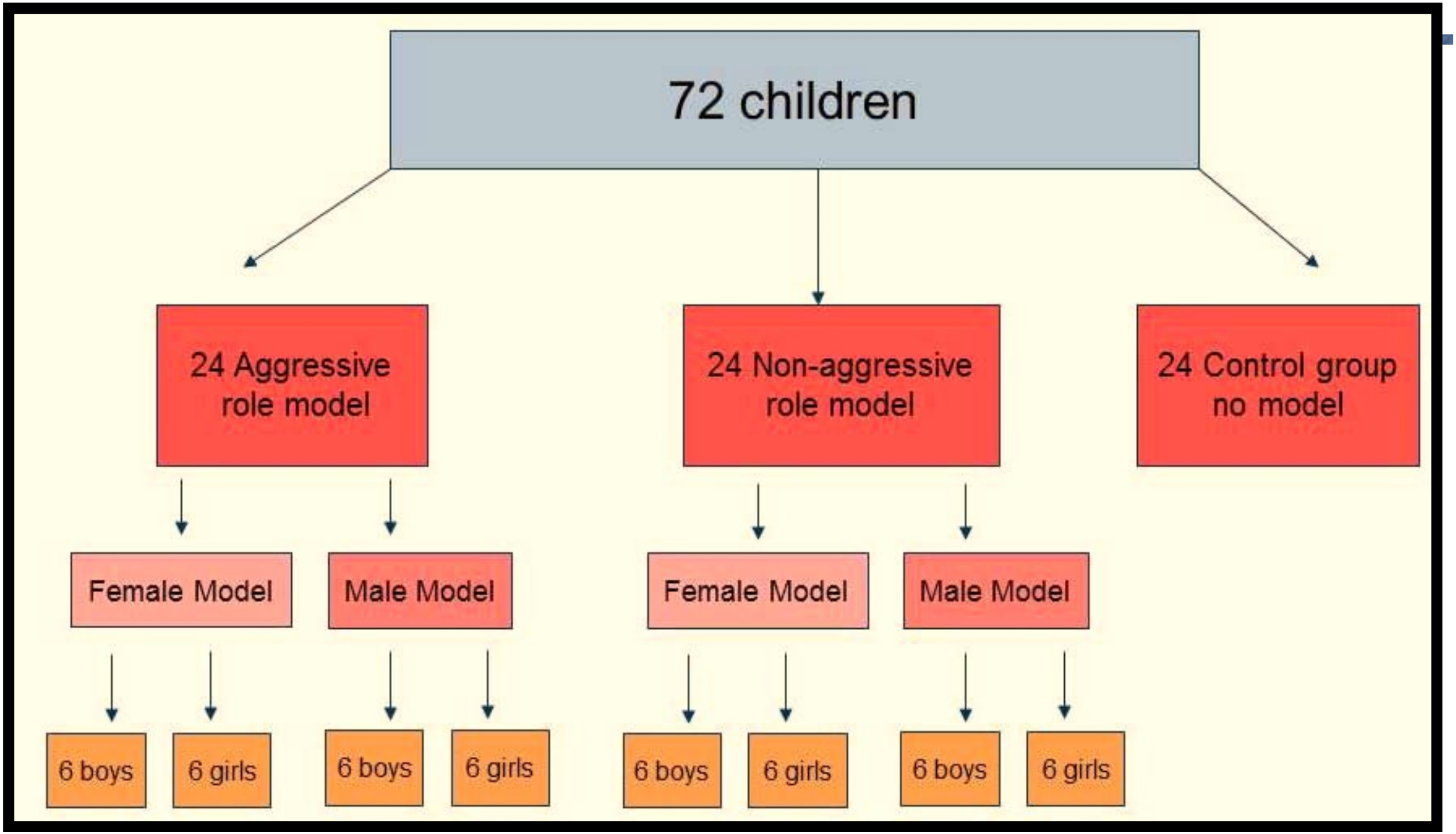
Encode

Imitate

Bobo Doll Experiment

- Bandura, Ross, and Ross (1961) tested 36 boys and 36 girls from the Stanford University Nursery School aged between 3 to 6 years old.
- The researchers pre-tested the children for how aggressive they were by observing the children in the nursery and judged their aggressive behavior on four 5-point rating scales.




72 children
24 Aggressive role model
24 Non-aggressive role model
24 Control group no model
Female Model
Male Model
Female Model
Male Model
6 boys
6 girls
6 boys
6 girls
6 boys
6 girls
6 boys
6 girls

Stage 1: Modeling

- 24 children (12 boys and 12 girls) watched a male or female model behaving aggressively towards a toy called a 'Bobo doll'. The adults attacked the Bobo doll in a distinctive manner - they used a hammer in some cases, and in others threw the doll in the air and shouted "Pow, Boom."
- Another 24 children (12 boys and 12 girls) were exposed to a non-aggressive model who played in a quiet and subdued manner for 10 minutes (playing with a tinker toy set and ignoring the bobo-doll).
- The final 24 children (12 boys and 12 girls) were used as a control group and not exposed to any model at all.



Stage 2: Aggression Arousal

- All the children (including the control group) were subjected to 'mild aggression arousal.' Each child was (separately) taken to a room with relatively attractive toys.
- As soon as the child started to play with the toys, the experimenter told the child that these were the experimenter's very best toys and she had decided to reserve them for the other children.

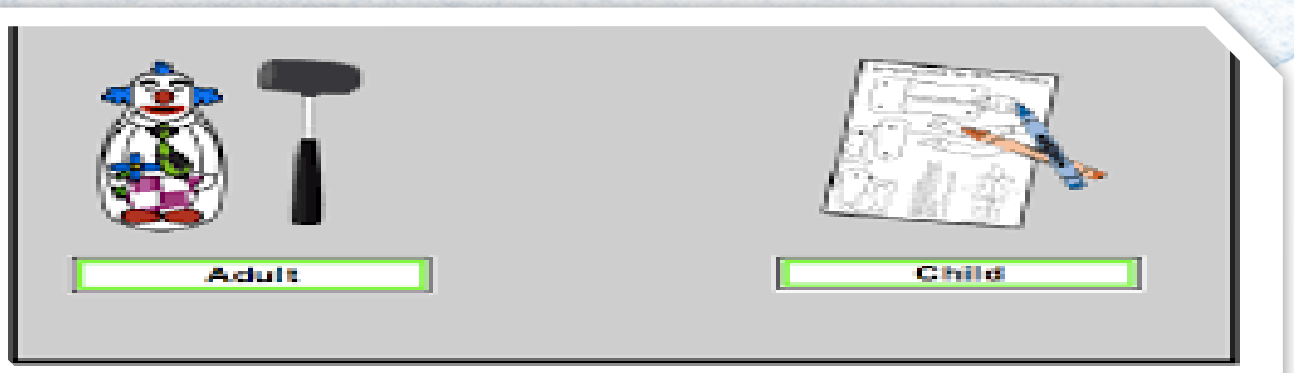
Stage 3: Test for Delayed Imitation

- The next room contained some aggressive toys and some non-aggressive toys
- The child was in the room for 20 minutes, and their behavior was observed and rated though a one-way mirror
- Other behaviors that didn't imitate that of the model were also recorded e.g., punching the Bobo doll on the nose.





RESULTS



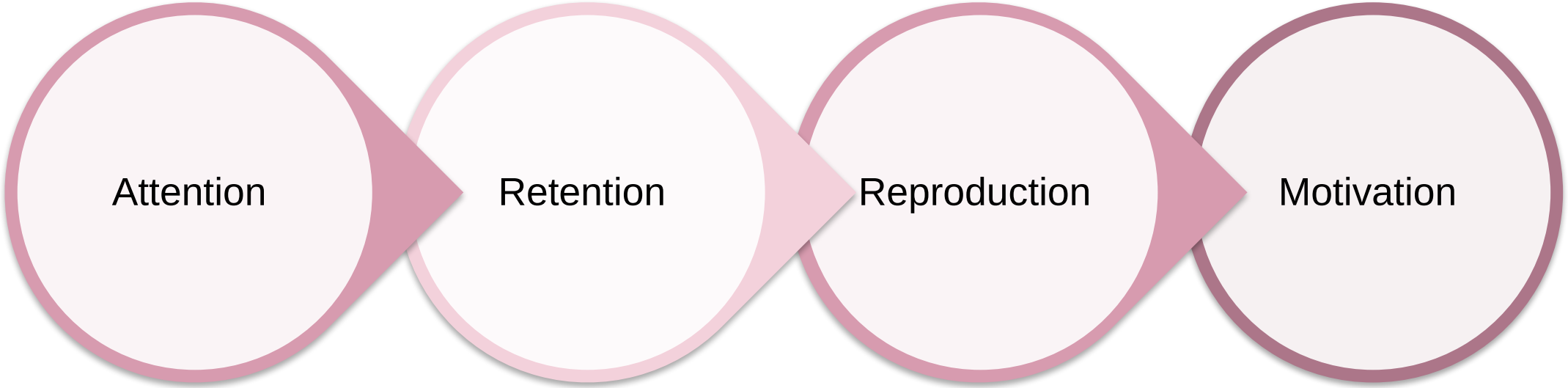


- Children who observed the aggressive model made far more imitative aggressive responses than those who were in the non-aggressive or control groups.
- The girls in the aggressive model condition also showed more physical aggressive responses if the model was male, but more verbal aggressive responses if the model was female. However, the exception to this general pattern was the observation of how often they punched Bobo, and in this case the effects of gender were reversed.



- Boys were more likely to imitate same-sex models than girls. The evidence for girls imitating same-sex models is not strong.
- Boys imitated more physically aggressive acts than girls. There was little difference in verbal aggression between boys and girls.

MECHANISM



- **Attention**: The individual needs to pay attention to the behavior and its consequences and form a mental representation of the behavior. For a behavior to be imitated, it has to grab our attention. Attention is therefore extremely important in whether a behavior influences others to imitating it.
- **Retention**: How well the behavior is ***remembered***. The behavior may be noticed but is it not always remembered which obviously prevents imitation. It is important therefore that a memory of the behavior is formed to be performed later by the observer.
- Much of social learning is not immediate, so this process is especially vital in those cases. Even if the behavior is reproduced shortly after seeing it, there needs to be a memory to refer to.



- **Reproduction**: This is the ability to perform the behavior that the model has just ***demonstrated***. We see much behavior on a daily basis that we would like to be able to imitate but that this not always possible. We are limited by our physical ability and for that reason, even if we wish to reproduce the behavior, we cannot.
- This influences our decisions whether to try and imitate it or not. Imagine the scenario of a 90-year-old-lady who struggles to walk watching Dancing on Ice. She may appreciate that the skill is a desirable one, but she will not attempt to imitate it because she physically cannot do it.



? **Motivation**: The will to perform the behavior. The rewards and punishment that follow a behavior will be considered by the observer. If the perceived rewards outweigh the perceived costs (if there are any), then the behavior will be more likely to be imitated by the observer. If the vicarious reinforcement is not seen to be important enough to the observer, then they will not imitate the behavior.



ASSIGNMENT

Q1. How Learning Can Affect Attitudes, Knowledge, or Behavior?

Q2. which can be the possible factors that affects learning??